



UAV target following in complex occluded environments with adaptive multi-modal fusion

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Abstract

Nowadays, deep reinforcement learning (DRL) has made remarkable achievements in unmanned aerial vehicle (UAV) target following. However, current DRL-based solutions only exploit appearance features to recognize and follow the target, and thus suffer from loss of the target in complex occluded environments. To that end, a novel target following solution based on adaptive appearance-motion feature fusion is proposed. First, we follow prior works to exploit convolutional neural network to extract appearance features of the target from the observation image captured from a down-looking camera. Meanwhile, we innovatively leverage action sequences of UAV to explicitly encode the motion features of the target. An attention module is subsequently introduced to adaptively select relevant useful features which serve as environment states and are fed into the decision module to produce the motion action of UAV. The whole network is trained using Deep Q-Network to learn the motion policy from observation in an end-to-end manner. We perform simulation experiments on Virtual Robot Experimentation Platform. Extensive experimental results demonstrate that: (1) Our proposed method achieves higher tracking accuracy and longer tracking time in various environments compared to state-of-the-art approaches; (2) The learned DRL policy could be well generalized to unseen environments.

Keywords UAV target following · Multi-modal fusion · Deep reinforcement learning · Complex occluded environments

1 Introduction

In the recent years, we have witnessed the rapid development of unmanned aerial vehicle (UAV) in military, aerial photography, surveillance and formation performance. In these applications, UAVs are typically required to automatically follow moving targets in order to fulfill various pre-defined tasks. As a consequence, UAV target following

has drawn more and more attention among researchers. At present, the task of target following can be divided into single-target following and multi-target following [1]. In this work, we mainly focus on single-target following.

UAV target following has been largely studied and many viable solutions have been proposed. Traditional approaches are typically divided into two categories according to the processing mode, one is based on generative models and the other is based on discriminative models. Generative models typically employ Kalman Filter [2] or Particle Filter [3] to estimate the states of the moving target (i.e., position, velocity, orientation and scale), and use the information obtained from the visual tracker to control the position and yaw angle of the UAV. These generative algorithms don't take the background information into account and only describe the target through a single mathematical model, which makes the accuracy of these algorithms relatively low. Different from generative models, discriminative models adopt a recognition method to discriminate the target and background from sequential images. Many state-of-the-art discriminative trackers are built upon correlation filter such as [4–9], and achieve exceptional performance gains by fully utilizing current frame information with a

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dense search. However, these discriminative trackers are relatively vulnerable to appearance variation and suffer from performance degradation in occluded environments. With the widespread application of deep learning (DL), experts consider applying it to object tracking [10–12]. These methods represented by SiamMask [11], first manually select the target in the initial image, use a convolutional neural network (CNN) for feature extraction, and then compare this feature with the features of each position in the subsequent frames to determine the target position. Due to the usage of image segmentation, the problem of partial occlusion can be solved to a certain extent. The studies of these visual tracking methods only focus on extracting 2D bounding boxes about targets from consecutive frames but do not consider how to control the movement of the agent. Therefore, these methods rely on human experts to manually label the target, and the perception system and control system are trained separately. End-to-end joint training of the perception layer and the control layer can improve the performance.

Deep reinforcement learning (DRL) implements end-to-end learning. DRL learns policies directly from raw observations through a trial-and-error process interacting with the environment, getting rid of the reliance on data labels. Recent advances in DRL offer new promises for solving the problem of UAV target following from the machine learning perspective. Li et al. [13] for the first time explore DRL technique combined with conventional feedback PID controller to learn UAV control practically. Li et al. [14] leverage DRL and meta-learning techniques to realize the control of UAV, allowing a UAV to track a target in an environment, where the motion of a target is uncertain. Bhagat et al. [15] develop a DRL technique with a curriculum training framework to track the target in the presence of obstacles and target motion uncertainty. These DRL-based solutions have proved to achieve comparable or better results compared to traditional methods. However, current DRL-based methods only exploit appearance features to recognize the target in an unsupervised manner, and thus suffer from loss of the target in complex occluded environments which are common in real-world applications.

To tackle the above challenges, a novel UAV target following solution based on adaptive appearance-motion feature fusion is proposed. First, we follow previous works [13] to exploit CNN to extract appearance features of the target from the raw observation image. Meanwhile, we innovatively leverage action sequences of the UAV to encode the motion features of the target. When the target is visible, appearance features are more informative for the UAV to detect the target. However, when the target is occluded, UAV needs to rely on the motion features to infer the position of the target. Based on the observations, we introduce an attention module to adaptively select relevant features according to the state of the target.

The selected features serve as hidden environment state and are fed into the decision module to produce the motion action of UAV. The whole network is trained using Deep Q-Network (DQN) to alleviate the dependency of our model on annotated samples. We perform simulation experiments on Virtual Robot Experimentation Platform (V-REP). Extensive experimental results under various simulated circumstances demonstrate that our proposed method achieves higher tracking accuracy and longer tracking time compared to state-of-the-art approaches. Specifically, our model maintains an average tracking time of 470s with the average relative distance being about 0.25m when the height of UAV is kept at 3m. Besides, experimental results verify that the learned policy could be well generalized to unseen environments with novel structures and textures. The main contributions of this paper are summarized as follows.

- We propose a novel UAV target following model based adaptive appearance-motion feature fusion, which is achieved by proposing a novel attention-based feature fusion model to assign different weights to each modality based on the current state of the target. This module has proved to improve target following performance in complex occluded environments.
- Unlike other methods that obtain motion features implicitly through image sequences, we encode motion features through action sequences to help the UAV follow the target effectively when occluded.
- Extensive experimental results demonstrate that our proposed method achieves higher tracking accuracy and longer tracking time compared to state-of-the-art approaches. Besides, experiment results verify that the learned policy could be well generalized to unseen environments with novel structures and textures.

The rest of the paper is organized as follows. Section 2 discusses the related background. Section 3 introduces our method and model. Section 4 describes and analyzes the experiments of UAV following a robot in different environments for a longtime. Section 5 concludes the paper.

2 Related work

Traditional following algorithm Early work of traditional methods mainly focused on the research of generative model following algorithms, such as Kalman Filter, Particle Filter, Meanshift algorithm, Camshift algorithm [1–3, 16–23]. These methods first establish a target model or extract target features and search for similar features in subsequent frames, then iterate step by step to achieve target positioning. In [3], the authors model the tracking target, and define a similarity metric to determine the matching

degree between the particles and the target. In the process of target searching, the method of Particle Filter will sprinkle some particles according to a certain distribution. The possible location of the target is determined by counting the similarity of these particles. At these positions, more new particles are added in the next frame to ensure that the target is tracked with a greater probability. [2, 18] complete a mission of an unmanned surface vehicle to track the target ship. Kim [18] uses a chasing control law and circling control law to achieve the task. In [19], an automatic spatio-temporal regularization framework for high-performance UAV tracking is proposed. These methods of generative algorithms do not take background into account, so some works use the discriminative following algorithm. Different from the general filter method for processing the background and target model, Huang et al. [6] optimizes the tracking algorithm on the basis of discriminant correlation filter and proposes aberrance repressed correlation filter. Zhang et al. [24] optimizes their tracking algorithm on the basis of kernel correlation filter tracker and proposes an irregular sampling method based on target contour band, which is the approximate contour area of the target. Li et al. [5] presents a novel long-term visual tracking algorithm FAST (Frequency tracker And Spatial detector). A well-trained object classifier is obtained by estimating the translation and scale and transforming the tracking model from the frequency domain to the spatial domain.

These traditional algorithms solve some tasks related to UAV target following. Kandath et al. [25], Zhang et al. [26], Chen et al. [27] and Wang et al. [28] let the UAV track the target in complex environments. In [26, 27], the target needs to avoid obstacles when it moves. Kandath et al. [25] uses UAV to provide information of environment for unmanned ground vehicle navigation. Wang et al. [28] proposes a framework for ground-air cooperative navigation in indoor environments to accomplish more complex tasks such as tracking indoor moving people. The authors utilize UAV to follow ground moving targets in a dynamic artificial potential field in [26]. At the same time, the UAV needs to avoid obstacles and the target may be blocked. In addition, cluttered environments are constructed for UAV following in [27] and the target may also be occluded. There are also many security applications using image and video automatic tracking to track and monitor in public areas [20]. However, the traditional algorithms have obvious shortcomings, that is, the background information of the image is not fully utilized and the appearance of the target itself has the characteristics of randomness and diversity. Therefore, describing the target to be tracked through a single mathematical model has many limitations.

More recently, many methods have been proposed to learn policies from observation. These methods can be divided into reinforcement learning (RL) and supervised

learning. With the widespread application of deep learning (DL) methods, DL has also begun to be used in target following [29, 30]. Trackers with multi-feature fusion and deep features have better following accuracy. In the first two years, the applications of DL in the field of target following was not smooth. Only the image data of the initial frame can be used, so there is a lack of a large amount of data for neural network learning. Only when researchers transfer the CNN trained on the classified image data set to target following, the target following method based on DL has been fully developed.

DRL-based tracking algorithm In recent years, DRL finds a variety of applications in navigation. In [31], the authors merge DRL with UAV navigation through massive multiple-input-multiple-output technique. Zhang et al. [32], Yuan et al. [33], Jaderberg et al. [34] and Nguyen et al. [35] use DRL for robot navigation. Yuan et al. [33] designs a multi-reward architecture for highway driving cars based on RL. The autonomous UAV achieves long-term trajectory path planning with DRL in [36]. The algorithm uses CNN to extract image features and learns a strategy function in the fully connected layer through RL. Then the algorithm optimizes the strategy function based on the target position in the previous frame and makes a decision based on the strategy function when the environmental state is known. Finally, accurate following is achieved. Benefiting from the good environment representation learned by the DRL, we intend to apply it to UAV target following. Li et al. [13] uses Deep Deterministic Policy Gradient to learn policies. The generated action is input to the low-level controller of UAV. The pid control method is used to promote the movement of UAV. Its mission is to follow moving pedestrians with the forward camera of UAV. In [37], the authors propose a coarse-tracker and a fine-tracker which are learned with an end-to-end DRL scheme to realize UAV following. However, the DRL-based method cannot make UAV effectively follow the target typically in the complex occluded environments. If the target is occluded by obstacles, it will cause the UAV to fail to follow the target.

In contrast to previous methods, we leverage both appearance and motion features of the target to improve UAV target following performance in complex occluded environments.

3 Proposed method

3.1 Problem formulation

In UAV target following phase, the next state of the agent (UAV) is determined only by the current state, therefore the UAV target following problem could be formulated as

a Markov Decision Process (MDP), which is described by a tuple $\mathcal{M} = \langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma \rangle$. Here, $\mathcal{S}$ is the set of information received by the agent from the environment, referred to as the state space. $\mathcal{A}$ is the set of selective actions of the agent, referred to as the action space. State transition probability is $\mathcal{P} : \mathcal{S} \times \mathcal{S} \times \mathcal{A} \rightarrow [0, 1]$, which represents the probability of performing a certain action and transferring from one state to another. $\mathcal{R} : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$ is the reward function, which is the feedback given by the environment based on the agent's action. $\gamma \in [0, 1]$ is the discount factor, which represents the importance of future returns relative to current returns. At each time step t , the environment is in state s_t . The agent selects an action $a_t \in \mathcal{A}$ according to a given policy $\pi : \mathcal{S} \rightarrow \mathcal{A}$, and receives a numerical reward r_{t+1} . The objective is to find an optimal policy π , that maximizes the expected sum of future discounted rewards, which is defined as follows.

$$V^\pi(s) = \mathbb{E}_\pi \left[\sum_{k=0}^{\infty} \gamma^k r_{t+k+1} | s_t = s \right]. \quad (1)$$

In our case, the environment state refers to the observation image obtained from the UAV-mounted down-looking camera. For the action space, we consider a set of high-level discrete actions, including *Forward*, *Back*, *Left*, *Right*, and *Stop*, as the output of the policy network, although the action space of UAVs is continuous in the real world. Introducing high-level discrete actions makes it much easier to transfer the learned policy to other UAV platforms.

We use DQN to solve the above MDP. DQN is an algorithm that combines DL and Q-Learning (QL) to realize end-to-end learning from observation to action. DQN no longer needs to record the Q value through a cumbersome Q table like QL and generates Q value through neural network. The update rule of neural network in DQN [38] is:

$$Q(s, a; \theta) \leftarrow Q(s, a; \theta) + \alpha[r + \gamma \max_{a'} Q(s', a'; \theta) - Q(s, a; \theta)], \quad (2)$$

where s means state and a means the action of the agent. The loss function of DQN is following:

$$L(\theta) = E \left[(Q_- - Q(s, a; \theta))^2 \right], \quad (3)$$

where $Q_- = r + \gamma \max_{a'} Q(s', a'; \theta)$. θ is the weight of network.

3.2 Network architecture

We present the overall framework of our appearance-motion fusion based target following architecture in Fig. 1. It

is composed of vision module, motion module, selective module and decision module. The vision module and motion module are designed to employ CNN and Long Short-Term Memory (LSTM) to extract appearance features and motion features of the target, respectively. The appearance and motion features are fused by the selective module, which adaptively select relevant features to help UAV infer the position of the target in the image space. The selective module is followed by the decision module, which generates actions of UAV based on the selected features.

3.2.1 Vision module

It leverages CNN to extract useful appearance features of the target from the observation image I_t captured from the UAV-mounted down-looking camera. The CNN architecture consists of three convolutional layers. Inspired by [13], we incorporate a spatial softmax layer and an expectation layer after the last convolutional layer in order to convert each pixel-wise feature map into one feature point in the image space. Let p_{ij}^c denote the value of pixel in row i and column j on c -th output feature map from the last convolutional layer. After spatial softmax layer, the pixel value will be updated as follows.

$$s_{ij}^c = \frac{e^{p_{ij}^c}}{\sum_{i'j'} e^{p_{i'j'}^c}}. \quad (4)$$

The following expectation layer calculates the expected image position of each feature map as follows.

$$(f_x^c, f_y^c) = \left(\sum_{ij} s_{ij}^c i, \sum_{ij} s_{ij}^c j \right). \quad (5)$$

It is observed that the expectation layer produces a set of spatial features that corresponds to points of maximal activation in the last convolutional layer. Here, each feature point could be regarded as one spatial estimate of the target in the image space. Finally, the C feature points obtained from the C feature maps are concatenated to construct the visual features of the target as follows.

$$\mathbf{f}_t^v = [f_x^1, f_y^1, f_x^2, f_y^2, \dots, f_x^C, f_y^C]. \quad (6)$$

3.2.2 Motion module

The scenes in real-world applications are typically complex and occluded. As a result, the targets are easily occluded

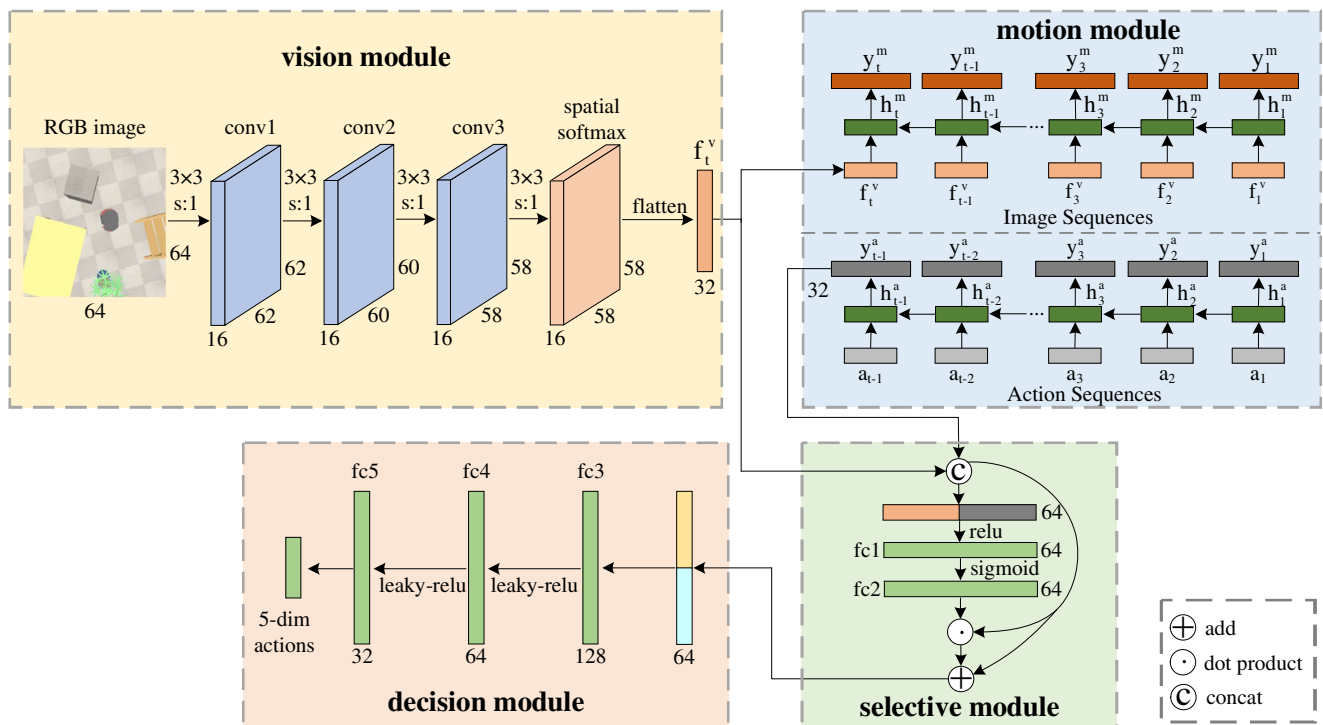


Fig. 1 Overall model architecture of our appearance-motion fusion based UAV target following framework, which takes the RGB observation image as inputs and generates the high-level discrete action

as outputs. The whole model is trained by DQN to alleviate the dependency of our method on labelled samples

during tracking process as shown in Fig. 2(c-d). In these scenarios, only using appearance features will lead to loss of targets, resulting in poor tracking performance. To alleviate this issue, we incorporate motion features to improve the target following performances in complex occluded environments.

Image sequences Recently, a lot of works employ LSTM to infer object motion directly from image sequences [39–41]. LSTM is a recurrent neural network that has a memory function, so it can correlate information on

time series and find the features in time series to make predictions. Inspired by this, we propose to employ LSTM to estimate the motion features of the target from a sequence of observed images. Here, the motion module is designed to share the same feature representation network with the vision module in order to reduce the model complexity of the whole tracking network. At each time step t , the inputs to LSTM include the spatial features f_t^v obtained at the current time t and the hidden state h_{t-1}^m of the recurrent network from previous time step $t - 1$, where m is a notation that refers to the feature is learned from the image



Fig. 2 For examples of observation images collected by UAV. (a) and (b) are the images where the target is visible. (c) and (d) are the images where the target is occluded. The target is in the blue bounding box

sequence. LSTM leverages three gate layers, including input gate, forget gate, and output gate, to control the flow of information and to achieve long-term memory. The memory cell of LSTM is implemented by the following equations:

$$i_t = \sigma(W_i \cdot [h_{t-1}^m, \mathbf{f}_t^v] + b_i), \quad (7)$$

$$f_t = \sigma(W_f \cdot [h_{t-1}^m, \mathbf{f}_t^v] + b_f), \quad (8)$$

$$\tilde{c}_t = \tanh(W_c \cdot [h_{t-1}^m, \mathbf{f}_t^v] + b_c), \quad (9)$$

$$c_t = f_t * c_{t-1} + i_t * \tilde{c}_t, \quad (10)$$

$$y_t^m = \sigma(W_y \cdot [h_{t-1}^m, \mathbf{f}_t^v] + b_y), \quad (11)$$

$$h_t^m = y_t^m * \tanh(c_t), \quad (12)$$

where σ is the logistic sigmoid function; i , f , y and c are the input gate, forget gate, output gate and cell vectors, respectively, all of them are same size as the hidden vector h^m . W_i , W_f , W_y , and W_c are learnable weight matrixes of the input gate, forget gate, output gate, and the cell, respectively. Similarly, b_i , b_f , b_y , and b_c refer to learnable weights bias of the input gate, forget gate, output gate, and the cell, respectively.

Action sequences During the process of UAV target following, UAV is encouraged to maintain a fixed relative position with respect to the target. As a consequence, the motion trajectory of UAV could reflect the motion features of the target to a certain extent. Based on the above observation, we propose an alternative method, which leverages action sequences of UAV to characterize the motion features of the target. Similarly, we employ LSTM to encode the action sequences. At each time step t , the inputs to LSTM include the encoded action a_t taken by UAV at current time t and the hidden state h_{t-1}^a of the recurrent network from the previous time step $t-1$, where a is a notation that refers to the feature is learned from the action sequence. Here, we utilize one-hot vector to encode discrete actions. Each action of *Forward*, *Back*, *Left*, *Right*, *Stop* is encoded by a 5-dimensional vector with a single '1' and the remainder '0's.

3.2.3 Selective module

Once appearance and motion features are extracted, the question following is how to fuse the two modalities

effectively. As shown in Fig. 2 (a-b), when the target is visible, the appearance features are more valuable to facilitate the UAV to recognize the target accurately in the image space. However, when the target is occluded as shown in Fig. 2(c-d), the appearance features are no longer available and the motion features become more important for UAV to infer the position of the target. The above observations suggest the necessity of fusing appearance-motion features adaptively for the purpose of guiding UAV to follow the target accurately under various scenarios. To that end, we incorporate an attention module to adaptively enhance useful features while suppressing irrelevant features. We concatenate the appearance features $\mathbf{f}_t^v$ and motion features $\mathbf{f}_t^m$ before feeding them into the attention module. Our attention module consists of two fully connected layers. Here, the fully connected layers are introduced to capture the cross-channel correlations. We regard the output vector A_t from the last fully connected layer as the weight vector, and dot-multiplies the input multi-modal feature vector $X_t = [\mathbf{f}_t^v, \mathbf{f}_t^m]$ by the weight vectors W_1 and W_2 to emphasize relevant channels. The final output of the attention block is obtained by adding the weighted feature map $\hat{X}_t$ to the original input feature map X_t . The implementation of the attention module could be described by the following equation.

$$\tilde{X}_t = A_t \cdot X_t + X_t = \sigma(W_2 \delta(W_1 X_t)) X_t + X_t, \quad (13)$$

where σ and δ refer to Sigmoid and ReLU activation functions, respectively; W_1 and W_2 denote the learnable parameters from the two fully connected layers, respectively.

3.2.4 Decision module

It receives the weighted feature map $\tilde{X}_t$ as the input and produces the action a_t of the UAV as the output. The decision module consists of three fully connected layers, with each layer followed by a Leaky ReLU function. The generated high-level action instruction is treated as the input to the bottom-level controller to control the movement of the low-level motors of UAV.

3.3 Reward shaping

The whole tracking model is trained in an end-to-end manner using the standard DQN loss function, which is described by (3). We can see that the value of $L(\theta)$ is closely related to the reward r . Actually, a good reward function could guide DQN faster towards more promising solutions. In this work, we employ the following reward function to encourage UAV to follow the target accurately.

$$r = e^{-\Delta pos}, \quad (14)$$

where $\Delta pos = \|pos_u - pos_t\|_2$ denotes the l_2 -norm Euclidean distance between the UAV and the target. The reward is designed to give UAV a positive reward, which is always positive and decreases with the relative distance, when the target is within the field of view (FOV) of UAV. Because the UAV has a fixed altitude during following, setting the reward in this way will gradually tend to the position of the UAV directly above the target. However, when the target is lost, the training period will be terminated.

4 Experiments

4.1 Simulator settings

4.1.1 Environments

In this paper, we conduct experiments on UAV target following in indoor environment to simulate a fully automatic unmanned warehouse environment [42, 43], where the target is a robot. We build the experimental environments by ourselves on V-REP, and show them in Fig. 3. These environments typically consist of ground robots, UAV, tables, chairs, stairs, wardrobes, bookcase, sofas, green plants, sand dunes and trees in order to simulate real indoor and outdoor environments. In all these environments, the UAV is expected to follow the moving ground robot accurately. The ground robot is equipped with 16 ultra-sonic sensors and it will rotate itself on a fixed angle when encountering obstacles, realizing automatic obstacle avoidance. When there are no obstacles, the robot's movement rule is as follows. The straight motion speed v_0 of the robot is set to 0.8m/s. Every 150 counts a scalar random number a between 0 and 1 is generated. If the value of a is smaller than 0.2, the robot's left wheel speed will increase by 0.5m/s and the robot will turn right. If this number is larger than 0.8, the speed of the right wheel is increased by 0.5m/s, and the robot will turn left. Otherwise, the robot will keep going straight. The moving procedure of the robot could be described in (15).

For the UAV, each generated high-level action lasts for 1.5s. In order to introduce dynamic elements, we also include multiple moving pedestrians in these environments. Their movement characteristics are described as follows: they keep going straight and rotate itself on a fixed angle only when they encounter an obstacle.

$$(v_L, v_R) = \begin{cases} (v_0 + 0.5, v_0), & a < 0.2 \\ (v_0, v_0), & 0.2 \leq a \leq 0.8 \\ (v_0, v_0 + 0.5), & a > 0.8 \end{cases}. \quad (15)$$

Dataset We generate a total of 9 scenes for training and testing our tracking model. We train our model in scene (a), and test it in scenes (a-i). Except for scene (a), the model has not been trained in other scenes and is directly tested, so it can reflect the generalization of the model. Scene (b) has similar appearances to scene (a), but includes more tables, leading to more serious occlusions. We use scenes (a-b) to test the tracking accuracy of our model. Scenes (c-e) have the similar textures to training scene (a), but they have different structures compared to scene (a). These three scenes are designed to test the generalization ability of our model on novel environments with different structures. It is worth noting that the difference between scene (c) and scene (d) is that a table is placed in front of the robot in (d), such that the robot will inevitably be occluded in this environment. Scenes (f-h) have different structures and different textures compared to scene (a), and these three scenes are designed to test the generalization capability of our model to novel environments on novel textures and structures. The floors in scenes (f-g) are concrete floors and scene (h) uses wooden floor. Scene (i) is a meeting room which has different structures and textures, and contains novel categories of objects compared to training scene (a).

4.1.2 Implementation details

Our experiments are based on V-REP. The model is trained with an initial exploration probability 0.9 to 0.1, which is reduced every step, 5000 steps in total. Finally, the exploration probability e remains 0.1 unchanged. We randomize a number between 0 and 1, if this number is less than e or the total number of steps is less than 400 steps, it will output a random action. Otherwise, the predictive action will be output through the model. The size of the input image is 64×64 . The vision module includes three convolutional layers and a spatial softmax layer. Leaky-relu function is used for nonlinear activation after every convolutional layers, except the sigmoid function in selective module. The discount factor γ is 0.99, batch-size is 64. All scenes use the quadrotor model and a pioneer 3dx robot in V-REP. The maximum number of episodes is set to 1500 during training. In the testing set, we design 100 episodes totally to calculate the average duration and average reward of each episode of seven algorithms in different environments. The simulation was implemented on the platform with an i7-9700K CPU, RTX2080Ti GPU, and 32GB of RAM.

4.1.3 Baselines and evaluation metrics

Traditional algorithms consist of trackers and control strategies. This paper considers two types of popular



Fig. 3 Nine scenes for training and testing our tracking model. We train our algorithms in scene (a), and test models in scenes (a-i). Scenes (c-e) have similar textures but different structures compared to scene (a). Scenes (f-i) have different textures and different structures compared to scene (a)

traditional algorithms, using Kernel Correlation Filter (KCF) [7] and SiamMask [11] as trackers respectively. The control strategy is based on [3], after obtaining the center point of the target and the size of the frame, the purpose is to make the UAV move just above the target. Then,

for the DRL algorithm, We compare our approach with the state-of-the-art method in [13], which employs DRL to learn the high-level policy directly from the observed image. We denote this model as **Vision Tracker** because it only exploits the appearance features of the target to infer

the position of the target. In addition, we also consider three variants of our model, including ViMFT, VaMFT, MA-VaMFT, as baselines for performance evaluation and comparison in order to prove the effectiveness of the proposed motion and selective module. Characteristics of each variant model are described as follows.

ViMFT is the model that extracts motion features of the target from a sequence of observed images, and leverages both appearance and motion features to follow the target. Different from our proposed model, ViMFT does not employ the selective module to fuse the appearance and motion features adaptively. Instead, it directly concatenates appearance and motion features and feed them into the decision module.

VaMFT shares the same structure with ViMFT except that it extracts the motion of the target from a sequence of actions from the UAV.

MA-VaMFT shares the same structure with our proposed method, except that it employs modal attention rather than channel attention to fuse the appearance and motion features. In Ma-VaMFT, the attention module outputs a weight vector A of length 2, and all channels from the same modality share the same weight.

In the following, we refer our proposed Channel-Attention based Vision-actionMotion Fusion Tracker as **CA-VaMFT** in order to facilitate explanation.

We follow [13] to employ the average duration and average reward as performance indicators to evaluate the performances of all these models. Average duration measures how long the UAV could follow the target accurately using a specific tracker. Here, a failures is regarded to occur when the target is out of FOV of the UAV. The average reward measures how much rewards the UAV receives on average within a fixed period. As the reward decreases with the relative distance between the UAV and target, the average reward actually reflects the tracking accuracy to a certain extent. At the same time, as real-time performance is very important in UAV target following, we also consider the inference efficiency of our model by counting the number of frames that the model can process per second, i.e., FPS.

4.2 Training results

We draw the reward curves of the five models trained with DRL in Fig. 4. We make the following observations from the reward curves: (1) First, our proposed model CA-VaMFT converges faster and finally converges to a better solution (i.e., average reward of about 0.82) compared to four other baselines; (2) Second, incorporating motion features

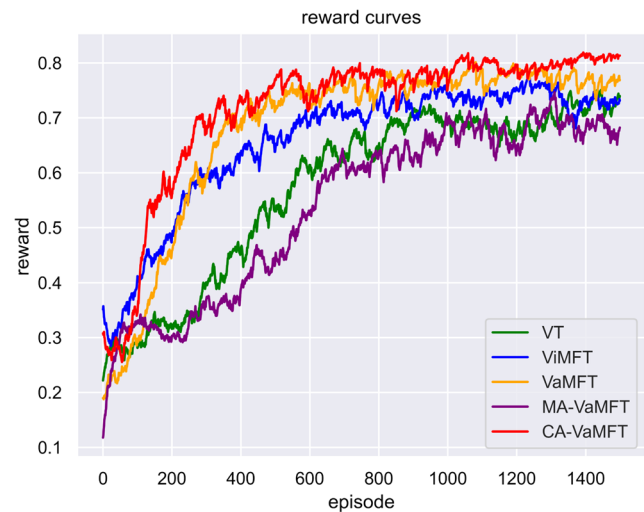


Fig. 4 Reward curves of five different models on the scene (a)

enable both ViMFT and VaMFT to reach better performance compared to VT, which suggests the importance of motion features in occluded environments; (3) Third, the superior performance of VaMFT to ViMFT proves that using action sequences to encode motion features is more effective; (4) Fourth, the performance of MA-VaMFT is surprisingly much worse than our proposed CA-VaMFT, which suggests the importance of fusing appearance and motion features in an effective way.

4.3 Numerical evaluation on tracking accuracy

We test all trained models in scenes (a-b) as shown in Fig. 3 to evaluate the tracking accuracy of all seven trackers. As mentioned before, scene (b) includes more tables, leading to more serious occlusions compared to scene (a). We report the average duration and average reward of each tracker in Tables 1 and 2, respectively. Here, we run each tracker 100 episodes in each period to calculate its average duration and average reward. When recording average duration, a period terminates only when the target is lost. However, the length of each period is fixed to 250s when recording the average reward of each tracker. The bold part emphasizes the performance of our method. We make the following observations from the two tables:

- (a) It can be seen from the tables that the tracking accuracy of traditional algorithms like KCF and SiamMask are not as good as our model. This observation can be explained that traditional visual following methods only focus on extracting bounding boxes about targets in consecutive frames, but do not consider how to

Table 1 Evaluation of all seven methods on average duration in various environments

Model \ Env	Scene (a)	Scene (b)	Scene (c)	Scene (d)	Scene (e)	Scene (f)	Scene (g)	Scene (h)	Scene (i)
KCF	113	98	93	86	83	90	82	80	35
SiamMask	180	152	132	112	105	128	109	101	37
VT	296	261	309	182	152	90	86	65	63
ViMFT	352	255	188	152	121	78	73	70	50
VaMFT	418	367	191	133	120	88	84	57	33
MA-VaMFT	312	216	248	99	97	80	69	51	26
CA-VaMFT	470	407	366	280	225	276	243	177	76

A total of 100 episodes to calculate the average duration of each episode, only when the relative distance between the target and agent is larger than $3.2m$, the episode will break

control the movement of the UAV. So end-to-end learning can improve the tracking performance.

- (b) Our proposed model CA-VaMFT achieves the best performance in both scenes in terms of both average duration and average reward. In particular, our model outperforms the second-performing VaMFT by a large margin in the scene (b) with more serious occlusions. This observation demonstrates the excellent tracking performances of our approach in occluded environments, and the same time verifies the effectiveness of incorporating motion features in an adaptive manner.
- (c) VaMFT outperforms both VT and ViMFT in both scenes in term of both average duration and average reward. On one hand, this observation suggests the importance of motion features in improving the UAV target following in occluded environments. On the other hand, this observation verifies the superiority of encoding motion features from action sequences.

This can be explained by the fact that it is usually challenging for CNN to accurately recognize the target in image space due to low texture of the target in low resolution.

- (d) The tracking performance of MA-VaMFT is inferior to all of CA-VaMFT, VaMFT, and ViMFT in both scenes. This observation again suggests the importance of fusing features in an effective manner. The poor performance of MA-VaMFT could be explained by the fact that applying modal attention to the concatenated features easily lead to the ignorance of features from the specific modality whose importance is relatively low.

4.4 Numerical evaluation on generalization

As mentioned above, the scenes in Fig. 3 (c-i) are quite different from the scene (a) used for training. Hence, we test the trained models on scenes (c-i) to evaluate the generalization capability of these models on unseen

Table 2 Evaluation of all seven methods on average reward in various environments

Model \ Env	Scene (a)	Scene (b)	Scene (c)	Scene (d)	Scene (e)	Scene (f)	Scene (g)	Scene (h)	Scene (i)
KCF	0.4470	0.4372	0.4343	0.4204	0.3742	0.3782	0.3525	0.3712	0.1852
SiamMask	0.4717	0.4533	0.4655	0.4242	0.3961	0.3911	0.3576	0.3893	0.1861
VT	0.7174	0.6523	0.7152	0.6136	0.3997	0.4665	0.4303	0.2567	0.2051
ViMFT	0.7608	0.7129	0.6926	0.5816	0.2885	0.2588	0.2538	0.2691	0.2053
VaMFT	0.7952	0.7236	0.6524	0.5313	0.2651	0.4278	0.4190	0.1654	0.2046
MA-VaMFT	0.6977	0.6560	0.5136	0.4096	0.3438	0.3079	0.2869	0.1604	0.1520
CA-VaMFT	0.7993	0.7703	0.7517	0.6216	0.4803	0.6672	0.5530	0.4123	0.2161

There are 100 episodes totally and each episode is limited within 250s to calculate the average reward of each episode, when the relative distance between the target and agent is larger than $3.2m$ or the limited time has reached, the episode will break

environments with novel structures or textures. We report the average duration and average reward of each tracker in Tables 1 and 2, respectively. We make the following observations:

- (a) The tracking performances of all models decrease with the increasing scene complexity. In detail, the tracking performances in scenes (c-e) are generally superior to that in scenes (f-h) due to the fact that the textures and structures of scenes (f-h) are both different from the training scene (a); all of these models produces the worst tracking performance in the most challenging scene (i) as it contains novel category of objects.
- (b) It can be seen from the tables that the generalization of traditional algorithms like KCF and SiamMask are not as good as our model. This observation can be explained that because the bounding box is slightly larger than the target, when there is a large change in the environment texture, it will lead to poor performance of the traditional algorithms. Meanwhile, with more occlusions, excessive changes from frame to frame may lead to certain errors in the predicted target area.
- (c) Although all of these seven methods suffer from obvious performance decrease in unseen scenes (c-i), our proposed CA-VaMFT still achieves the best tracking performances in all these seven scenes. This observation verifies the strong generalization capability of our approach on unseen environments with novel textures or structures. The excellent generalization of CA-VaMFT might be due to the incorporation of the channel attention to focus on relevant features and to fuse the appearance features and the motion features intelligently.
- (d) Different from the training results, VT surprisingly outperforms both VaMFT and ViMFT on average in these six unseen scenes. The relative poor performances of VaMFT and ViMFT on unseen environments could be explained by the increased model complexity within our decision module, which is caused by the increased dimensionality of the input features, which concatenate appearance and motion features.
- (e) Although VaMFT performs slightly inferior to ViMFT on scenes (c-e), VaMFT outperforms ViMFT by a large margin on scenes (f-g). We explain the huge performance improvement by the following facts. As mentioned before, these two scenes share different textures with the training scene (a), which poses a challenge to our image-based motion module. In such novel scenes, our action-based motion module is more effective to infer the moving trajectory of the target from UAV action sequences. Besides, the relative poor performance of VaMFT on scene (h) might be due to novel structure of the scene, which makes the motion of the target less predictable.
- (f) The performance of MA-VaMFT is inferior to that of CA-VaMFT on average in all these unseen environments. This observation further verifies the effectiveness of our channel attention based multi-modal fusion module.

4.5 Numerical evaluation on varying occlusion degree

We design two new testing environments in Fig. 5(b) and (c) to test the effectiveness of our model under different degrees of occlusion, corresponding to scene (j) and scene (k) in Tables 3 and 4, respectively. The bold part emphasizes the performance of our method. The structures and textures

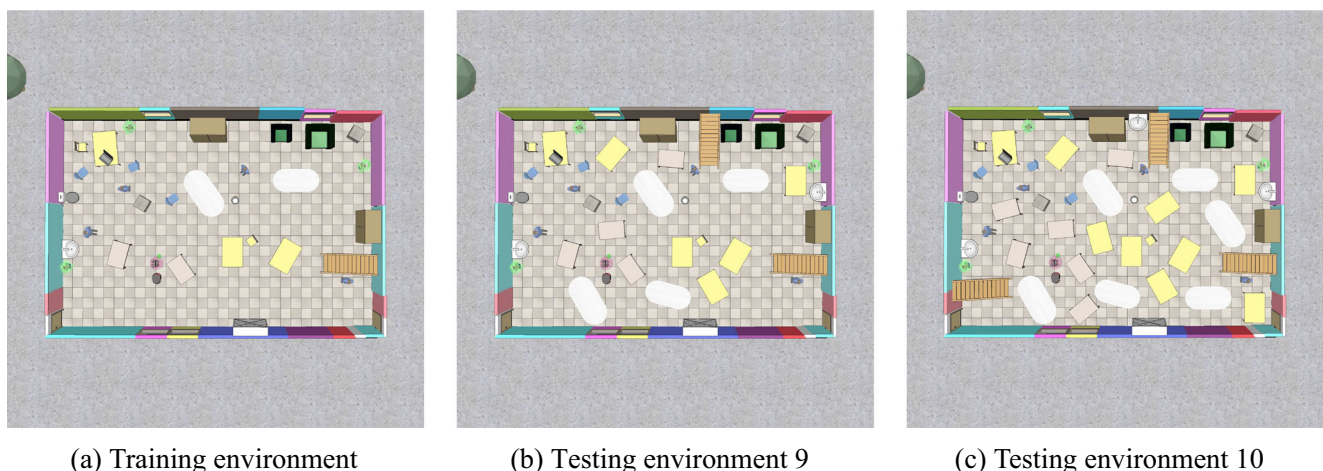


Fig. 5 Three environments with different occlusion degrees are designed to test the effectiveness of our model. These three environments have multiple occlusions such as tables, coffee tables, and ladders. The occlusion rates are 10%, 20%, and 30%, respectively

Table 3 Evaluation of all seven methods on average duration in three environments

Model Env	KCF	SiamMask	VT	ViMFT	VaMFT	MA-VaMFT	CA-VaMFT
Scene (a)	113	180	296	352	418	312	470
Scene (j)	103	163	276	196	213	279	412
Scene (k)	95	143	240	181	183	247	363

Only when the relative distance between the target and agent is larger than $3.2m$, the episode will break

of these three environments are similar, and the occlusion rates are increased by increasing the number of occluders. The occlusion rates of scene (j) and scene (k) are double and triple that of scene (a), i.e. 20% and 30%, respectively. Similarly, we report the average duration and average reward of each tracker in Tables 3 and 4, respectively. Here, we run each tracker 100 episodes in each period to calculate its average duration and average reward. When recording average duration, a period terminates only when the target is lost. However, the length of each period is fixed to 250s when recording the average reward of each tracker.

By comparing the performance of these algorithms in Tables 3 and 4 on scene (a), scene (j), and scene (k), it can be seen that although the performance of each algorithm decreases as the occlusion degree multiplies increases, CA-VaMFT outperforms other algorithms because it intelligently fuses appearance features and motion features through an attention mechanism, which enables our method to achieve the long-term effective following in environments with high occlusion rates.

4.6 Numerical evaluation on model inference speed

Because the real-time performance of the UAV is crucial in the task of target following, we compare our algorithm with other baselines and calculate the average decision time in

all scenes to obtain frames per second (FPS), as shown in Table 5.

We can see from the table that the FPS of KCF is large but the tracking accuracy is low. Due to the large amount of computations involved in deep neural network architecture, SiamMask achieves the lowest FPS. In comparison, our method processes 23 frames per second, which could be further accelerated by parallel computing and model compression.

4.7 Visualization evaluations

4.7.1 Visualization on feature representations

The spatial softmax layer within the proposed vision module is designed to produce C feature points. Each feature point is regarded as one spatial estimate of the target. We present four examples to show the outputs of the spatial softmax layer in Fig. 6, where each blue circle represents one output feature point. It is observed that all output feature points are distributed around the center of the robot in all four examples. This observation proves that the spatial softmax layer can effectively infer the position of the target in the image to a certain extent. And we can see from Fig. 6(d) that despite the similar shapes and colors of toilet and robot, our method is able to identify and track the correct target.

Table 4 Evaluation of all seven methods on average reward in three environments

Model Env	KCF	SiamMask	VT	ViMFT	VaMFT	MA-VaMFT	CA-VaMFT
Scene (a)	0.4470	0.4717	0.7174	0.7608	0.7952	0.6977	0.7993
Scene (j)	0.4119	0.4592	0.6538	0.6587	0.7340	0.5594	0.7416
Scene (k)	0.3844	0.4487	0.6109	0.6351	0.6741	0.5299	0.7157

Each episode is limited within 250s to calculate the average reward of each episode, when the relative distance between the target and agent is larger than $3.2m$ or the limited time has reached, the episode will break

Table 5 Evaluation of all seven methods on average FPS in various environments

Model	KCF	SiamMask	VT	ViMFT	VaMFT	MA-VaMFT	CA-VaMFT
FPS	68	12	27	25	24	20	23

4.7.2 Visualization on attention vectors

We present four examples to visualize the attention vectors from the selective module in Fig. 7. Figure 7(a) and (c) are the observed images where the robot is visible. In contrast, Fig. 7(e) and (g) are the observed images where the robot is occluded by the stair and table. The generated attention vectors are shown in Fig. 7(b), (d), (f) and (h), respectively. Here, each weight vector is of length 64, where the first 32 dimensions refer to the weights assigned to the appearance channels and the last 32 dimensions denote the weights assigned to the motion channels. We make the following observations: (1) In case (a), the average weight of first 32 and the last 32 dimensions are 0.71 and 0.63, respectively, and the maximum weight is assigned to twenty-third dimension; Similarly in case (c), the average weight of first 32 and the last 32 dimensions are 0.71 and 0.64, respectively, and the maximum weight is assigned to twenty-eighth dimension. This observation suggests that our tracking model pays more attention to the appearance features in order to infer the position of the target when the target is visible; (2) In case (e), the average weight of first 32 and the last 32 dimensions are 0.62 and 0.67, respectively, and the maximum weight is assigned to sixty-second dimension; Similarly in case (g), the average weight of first 32 and the last 32 dimensions are 0.61 and 0.67, respectively, and the maximum weight is assigned to sixty-second dimension. This observation suggests that the motion features are considered to be more important to infer the position of the target by our tracking model our when the target is occluded. The above observations

demonstrate the capability of our tracking model to adaptively fuse appearance-motion features according to the state of the target, which in turn explains the excellent tracking performance of our model.

4.7.3 Visualization on following paths

We run our tracker in four different environments, and visualize the learned trajectory of the UAV in Figs. 8 and 9. In these figures, the red and blue curves represent the traveling paths of the UAV and the robot, respectively; the markers 'o' and 'x' represent the starting point and the end point of the robot, respectively. We consider two set of experiments. In the first set of experiments, the episode is limited to 250s. Once the target is lost or the limited time is reached, the episode will terminate. The learned traveling paths of the UAV in these four scenes are displayed in Fig. 8(a-d). We observe that the UAV follows the target successfully across the whole period in all four scenes, and the relative distances between the UAV and the targets are within 0.25m on average. In the second set of experiments, the episode terminates only when the target is lost. We display the learned traveling paths of the UAV under such setting in Fig. 9(a-d). It is observed that the UAV follows the target successfully when the target suffers from a short period of occlusion. The loss of the target occurs when the target is occluded by the desks, tea tables or the sink for a certain long period. The loss of the target under long-term occlusion can be explained by the fact that our action module fails to predict the position of the target due to lack of informative inputs.

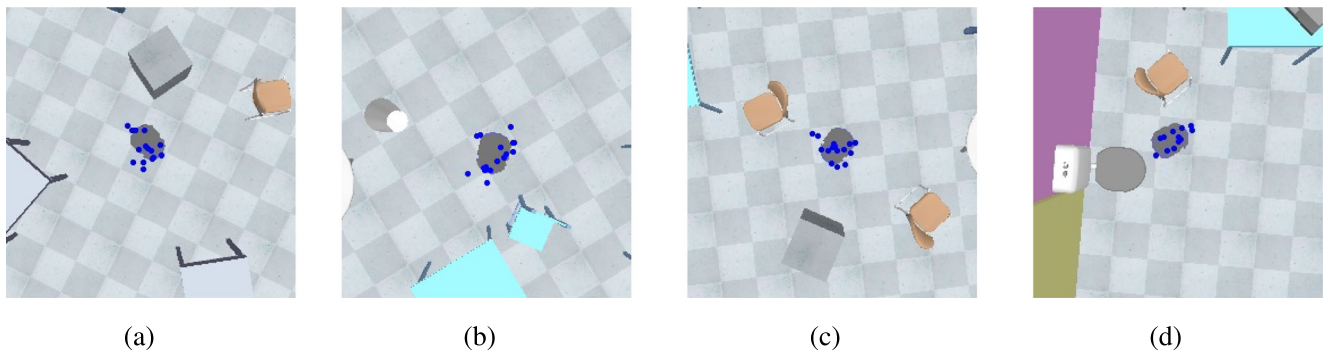


Fig. 6 Four examples to show the outputs of the spatial softmax layer within the vision module, where each blue circle represents one output feature point

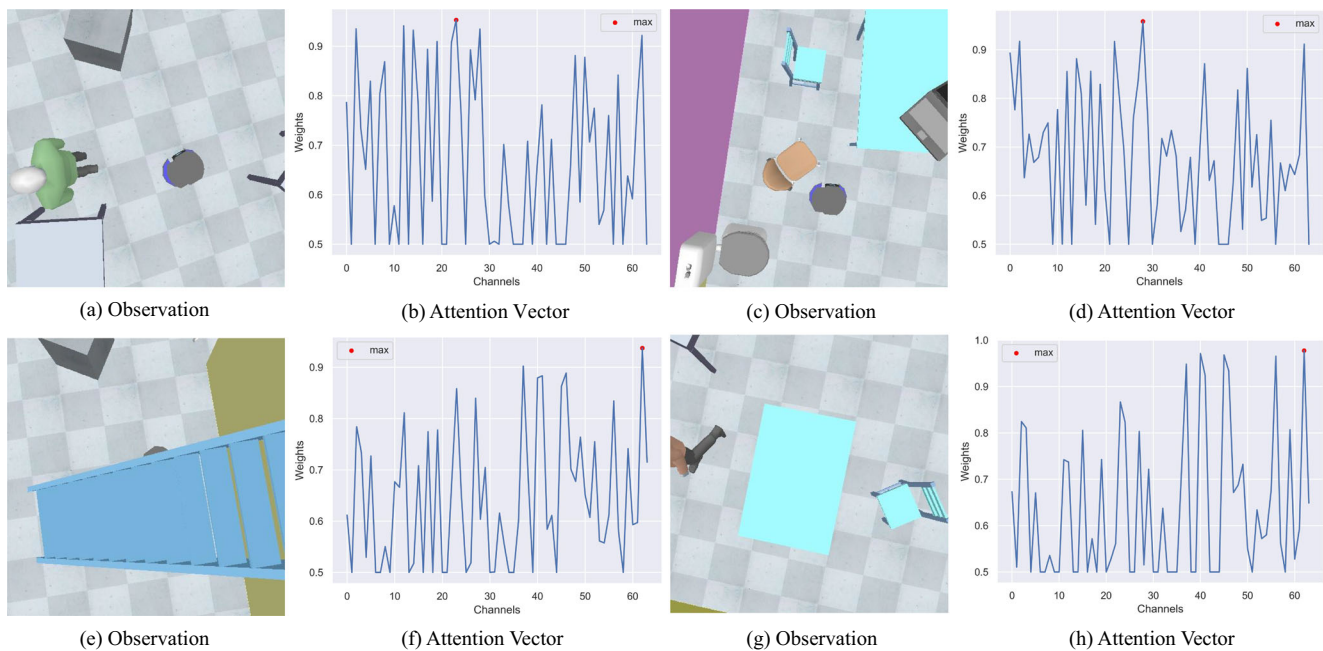


Fig. 7 Four examples to visualize the generated weight vectors from our selective module. Here, each weight vector is of length 64, where the first 32 dimensions refer to the weights assigned to the appearance channels and the last 32 dimensions denote the weights assigned to the motion channels

5 Conclusion

In this paper, a novel target tracking solution based on adaptive appearance-motion feature fusion is proposed to tackle the task of UAV target following in complex occluded environments. The whole network is trained using DQN to learn the motion policy from observation in an end-to-end manner. We perform simulation experiments on V-REP with our proposed method, and test it in several different complex occluded environments. These environments may have different floors or different furniture. Experimental results demonstrate that our algorithm outperforms the state-of-the-art DRL solutions. We test in the training environment at a height of $3m$, the episode will terminate

when the target is not in the FOV of UAV. Under the above conditions, we maintain an average tracking time of 470s every episode, and the average relative distance is kept at about $0.25m$. Besides, experiment results verify that our method achieves good generalization in other environments including mazes and meeting rooms with different floors or items. Next, we would like to consider the change of scale so that the UAV can follow the target smoothly at different heights. Secondly, although our method can be extended to outdoors, we do not take into account factors such as illumination changes. We will increase the outdoor UAV target following scenarios in later research, such as UAV rescue after disasters, UAV wild animals tracking, etc.

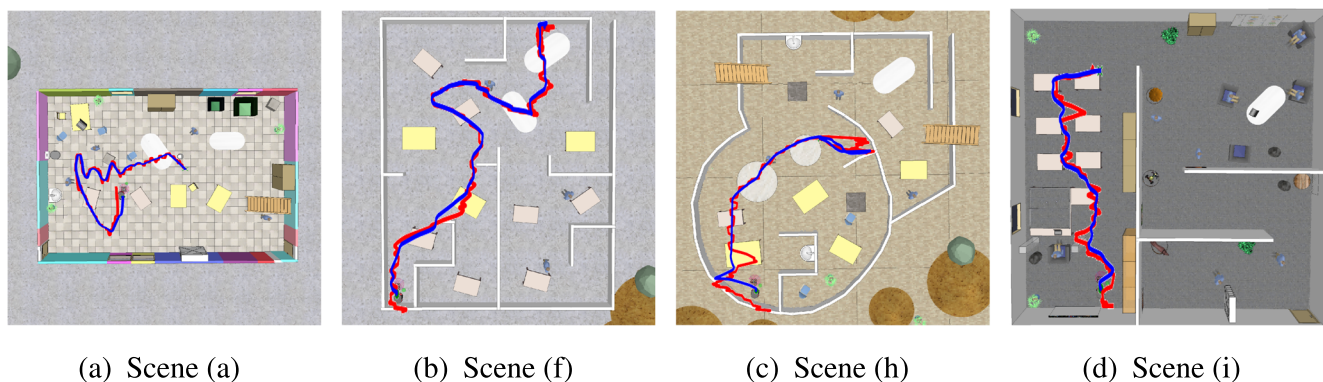


Fig. 8 Four examples to visualize the path of UAV and the robot when the episode time is limited to 250s

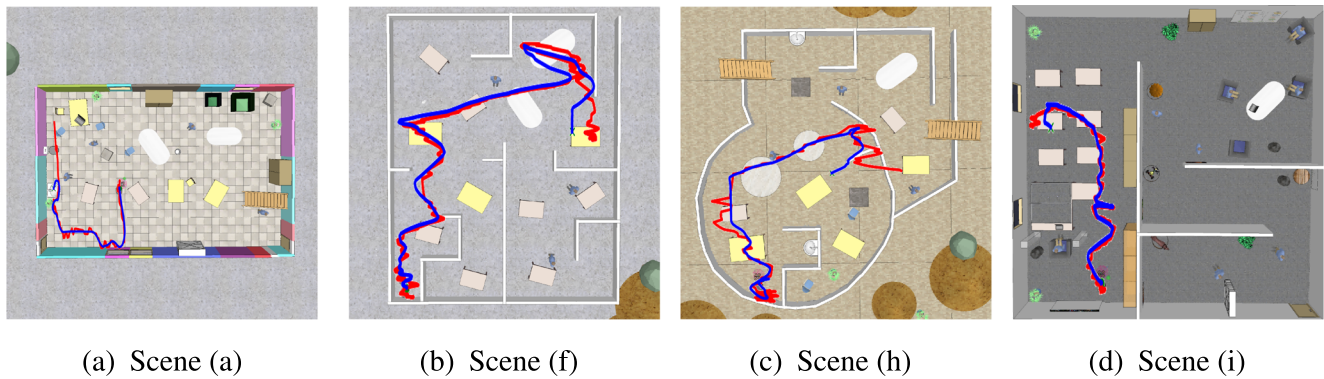


Fig. 9 Four examples to visualize the path of UAV and the robot when the episode time is not limited

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